Digital Image Processing

Lecture # 2
Fundamentals

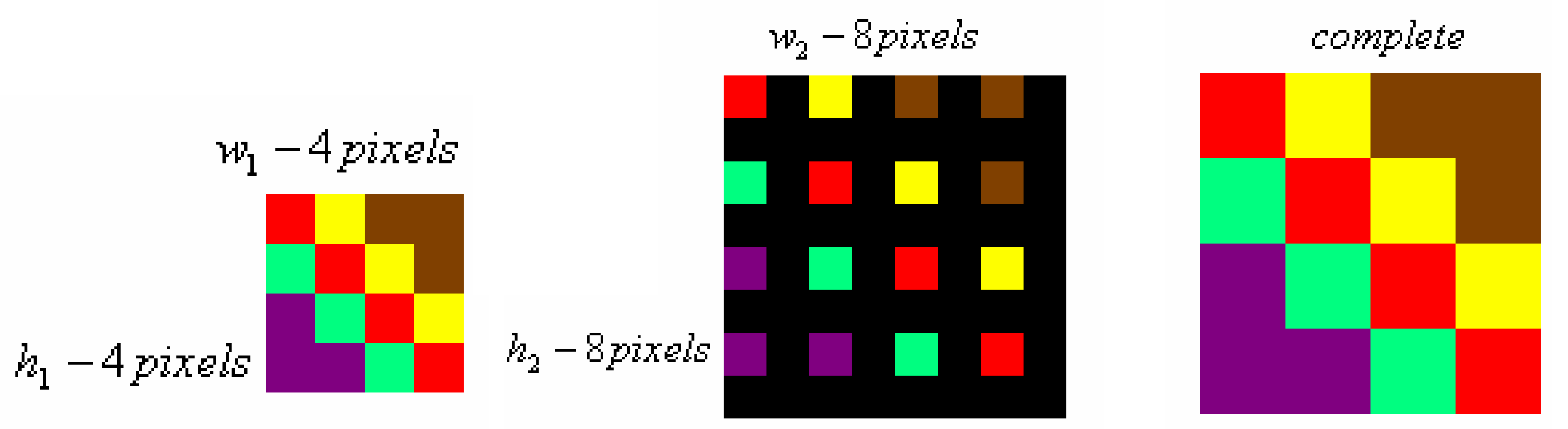
Image Resizing

- Pixel replication

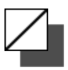
[1 2 3 4 5]

[1 1 2 2 3 3 4 4 5 5]   (One step)

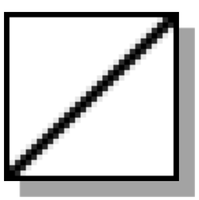
[1 1 1 2 2 2 3 3 3 4 4 4 5 5 5] (Two step)



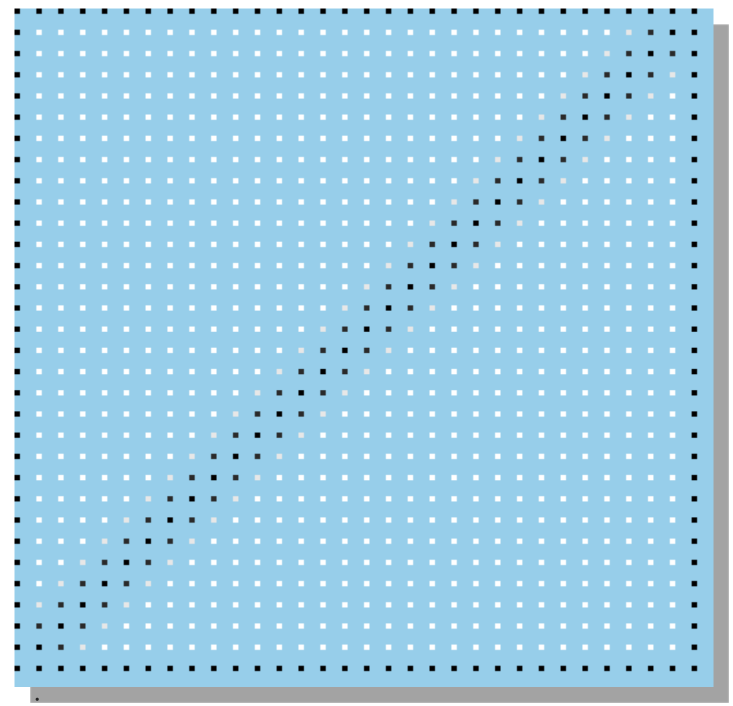
Enlarging an Image



Example: zoom this image 4x to get this image.

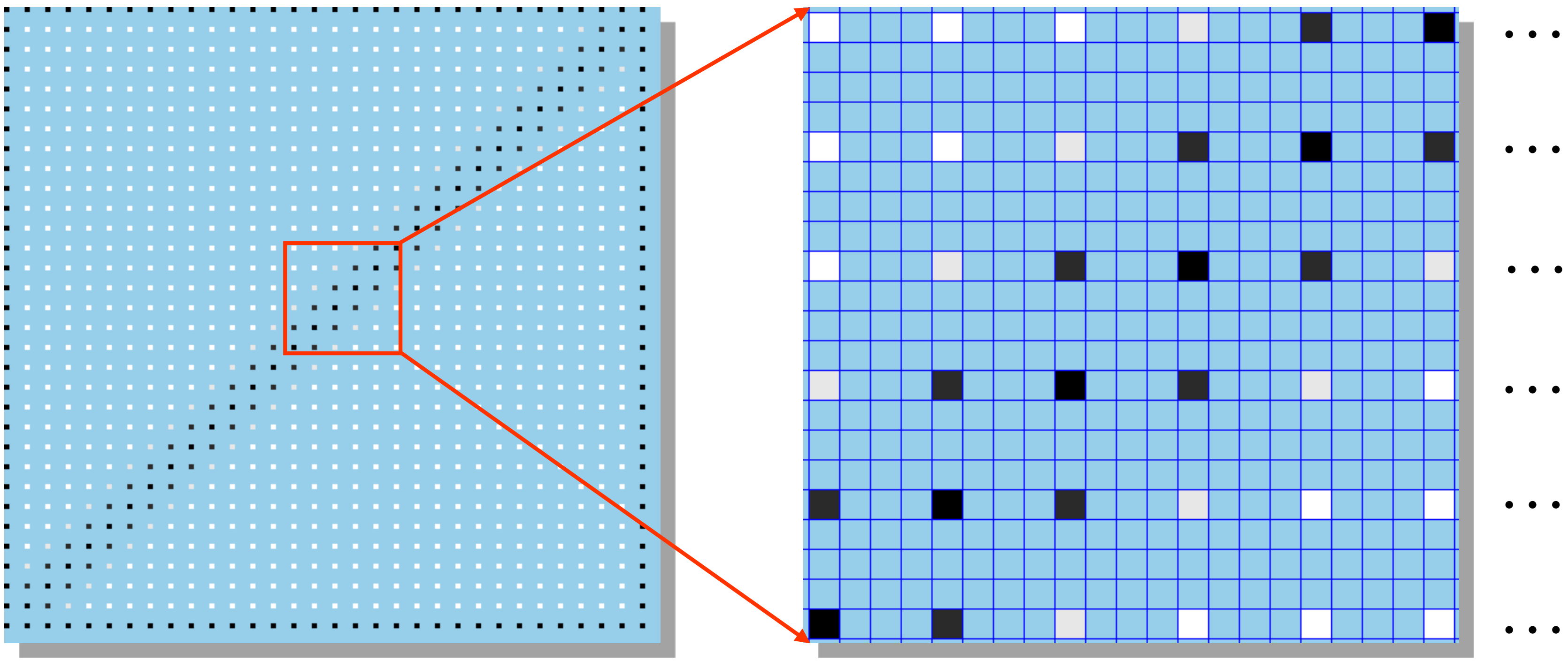


Start with a blank image 4 times the linear dimensions of the original.



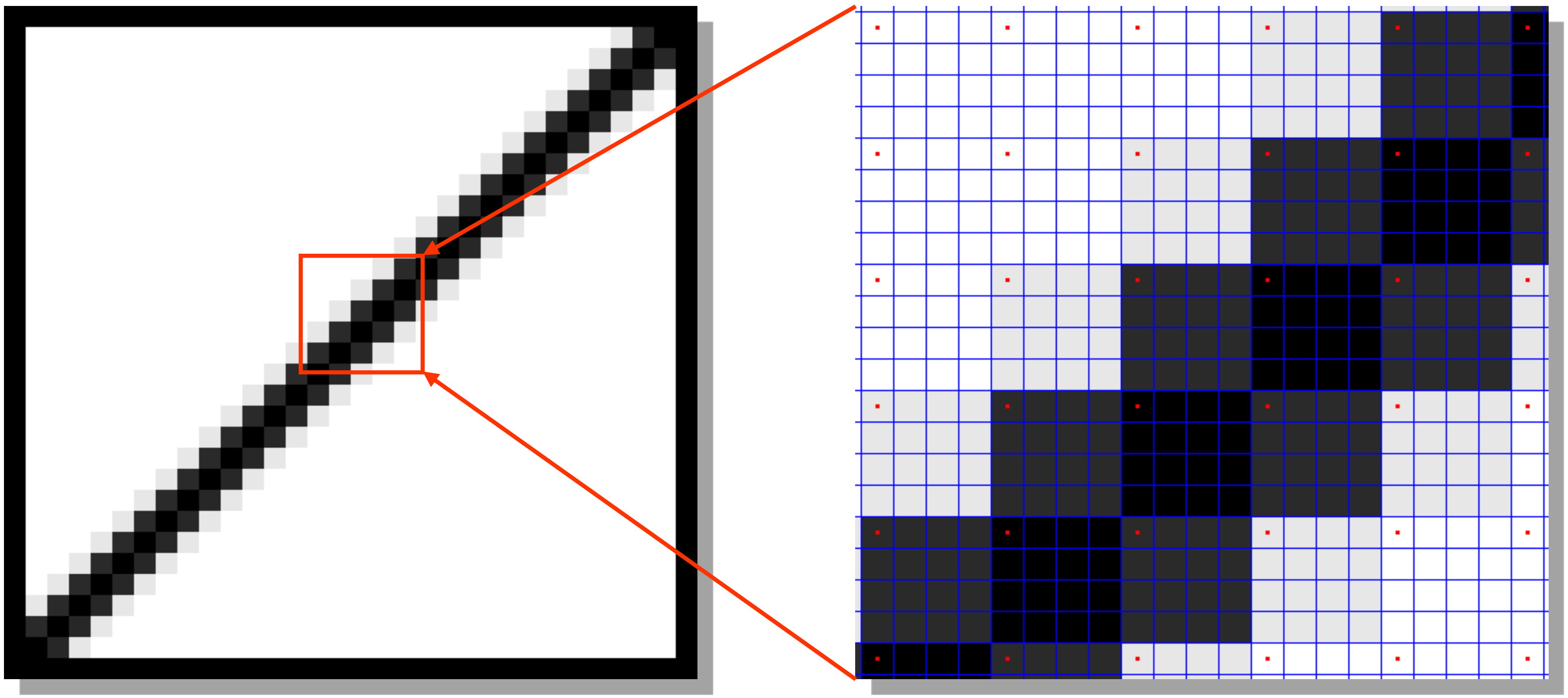
Fill in every 4th pixel in every 4th row with the original pixel values.

Enlarging an Image



Detail showing every 4th pixel in every 4th row with the original pixel values.

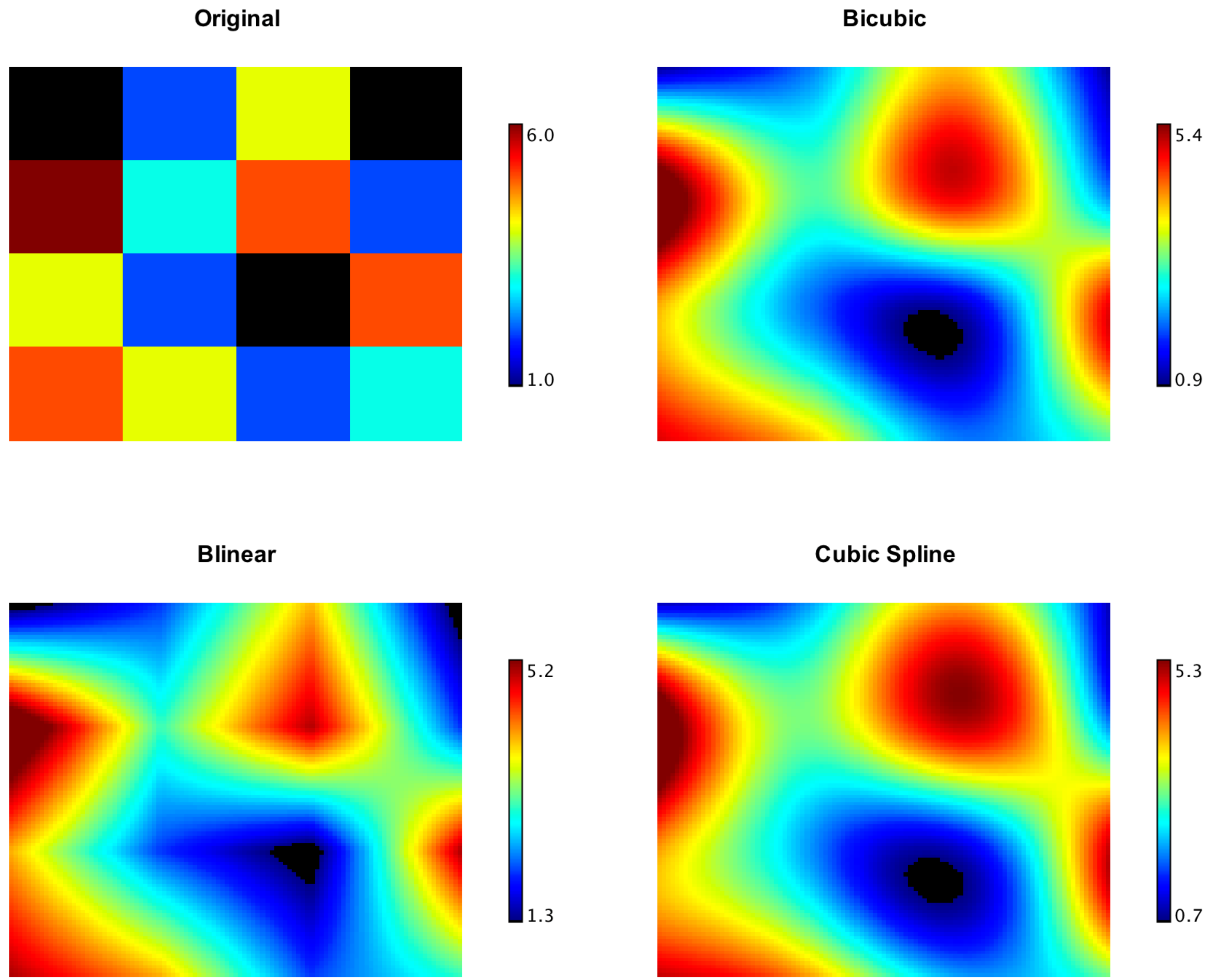
Enlarging an Image



Replicate the values

Image Interpolation

- Nearest neighbour interpolation
  - Simple but produces undesired artefacts
- Bilinear Interpolation
  - Contribution from 4 neighbors
- Bicubic Interpolation
  - Contribution from 16 neighbors



Original
6.0
1.0
Bicubic
5.4
0.9
Blinear
5.2
1.3
Cubic Spline
5.3
0.7

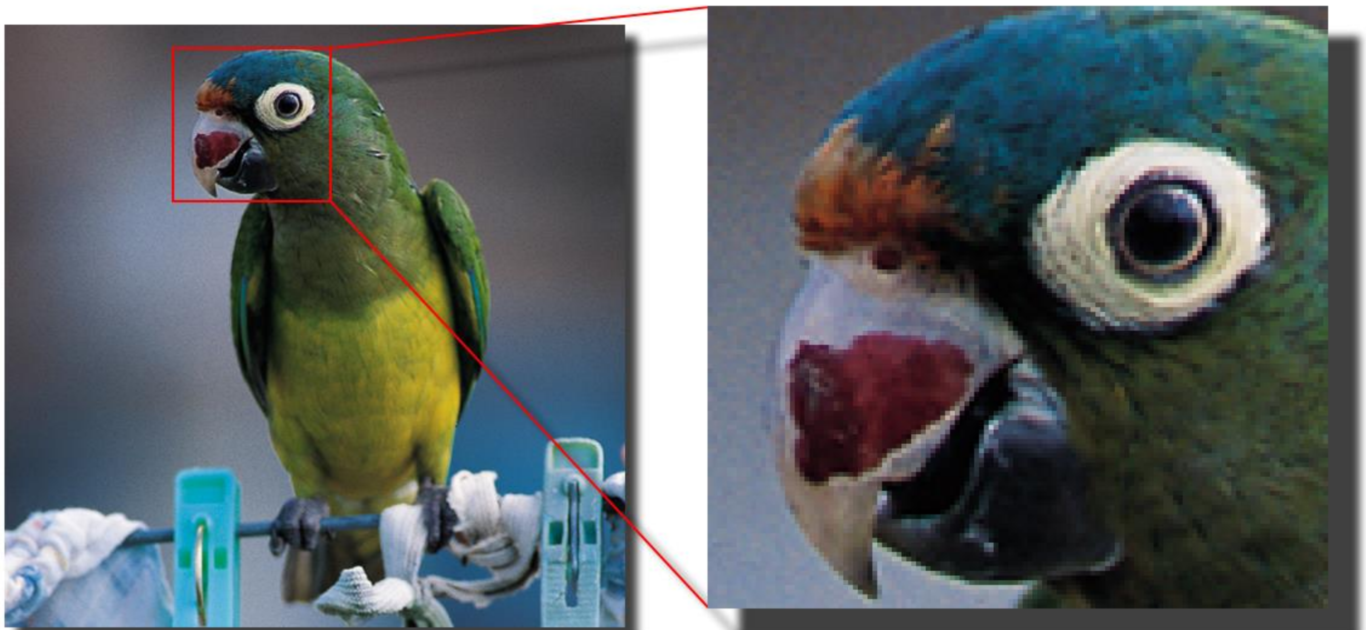
Interpolation: Comparison

We'll enlarge this image by a factor of 4 …

… via bilinear interpolation and compare it to a nearest neighbor enlargement.

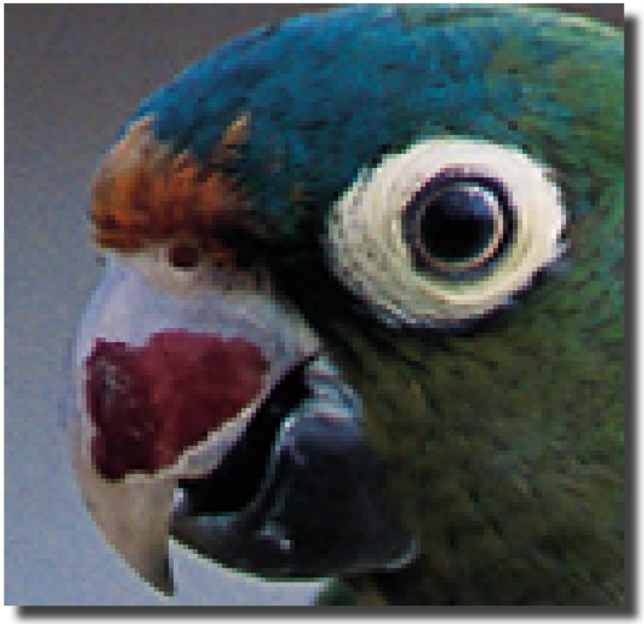
Interpolation: Comparison

Original Image



To better see what happens, we'll look at the parrot's eye.

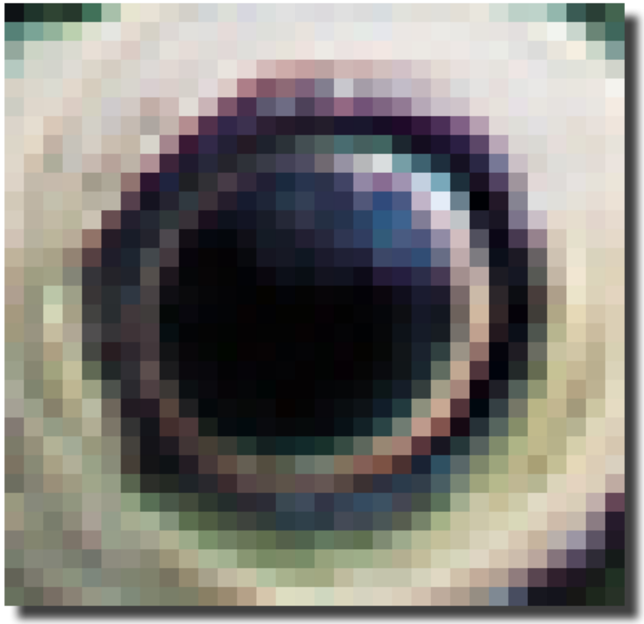
Interpolation: Comparison



Pixel replication

Bilinear interpolation

Interpolation: Comparison

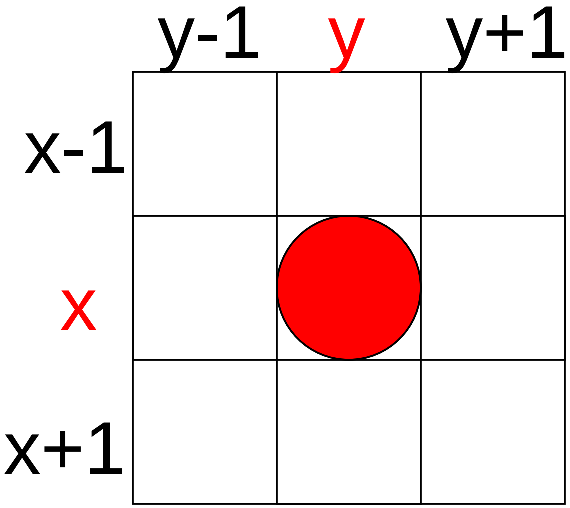


Pixel replication

Bilinear interpolation

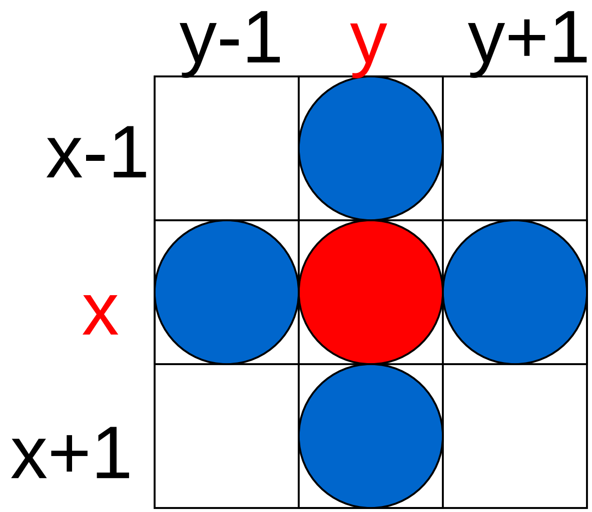
Relationships between pixels

- Neighbors of pixel are the pixels that are adjacent pixels of an identified pixel



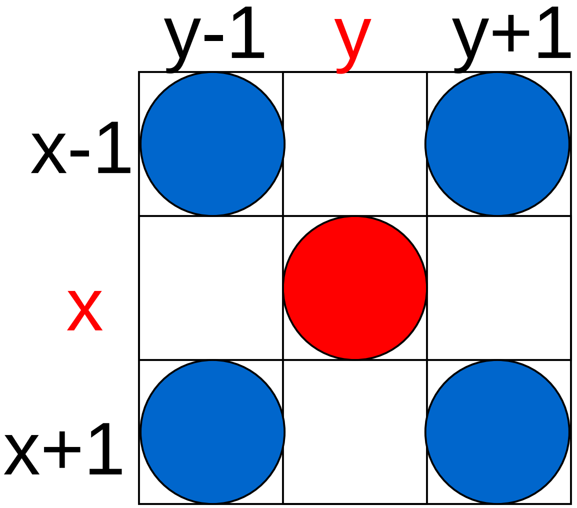
4- Neighbors of a Pixel –$N_4(p)$

What are the coordinates of each of the blue pixels



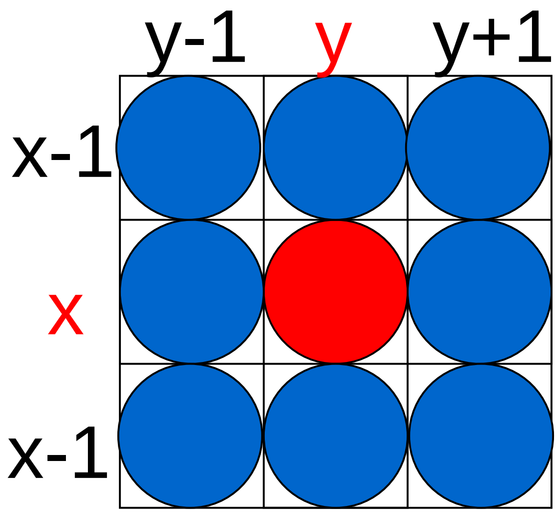
(x-1,y), (x+1,y), (x, y-1), (x, y+1)

Diagonal Neighbors of a Pixel –$N_D(p)$



(x-1,y-1), (x+1,y-1), (x-1, y+1), (x+1, y+1)

8- Neighbors of a Pixel –N_8(p)

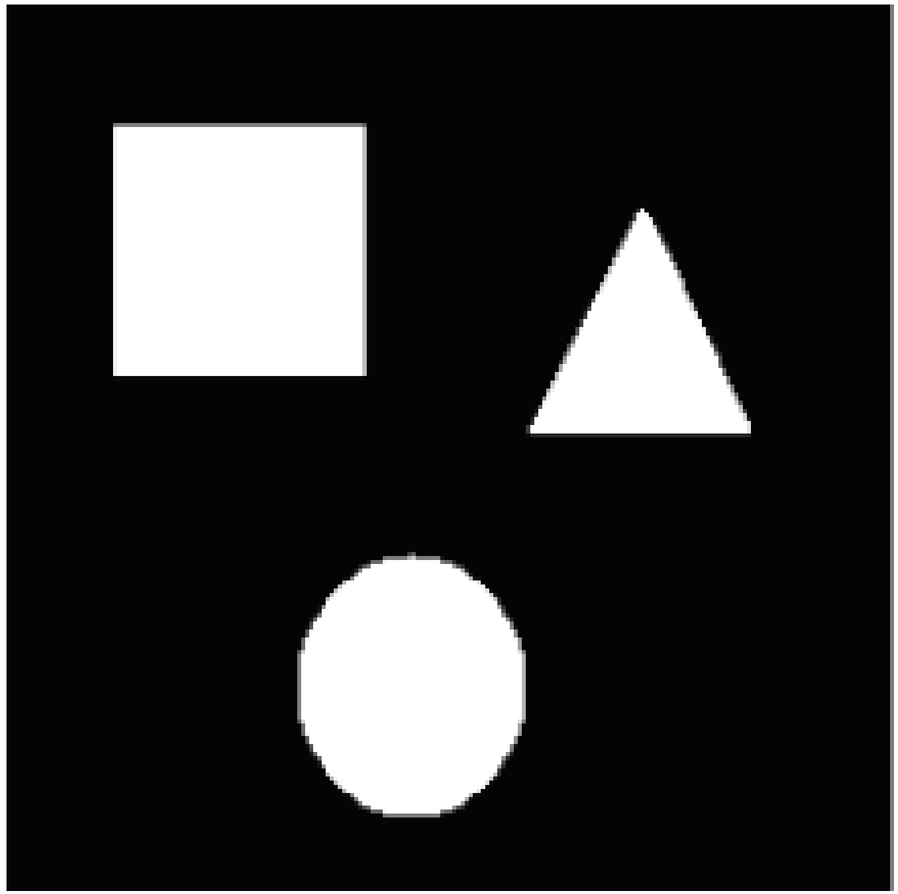


$$N_8(p) = N_4(p) \cup N_D(p)$$

(x-1,y), (x+1,y), (x, y-1), (x, y+1)

(x-1,y-1), (x+1,y-1), (x-1, y+1), (x+1, y+1)

Determine different regions in the image



Connectivity

- Establishing boundaries of objects and components in an image
- Group the same region by assumption that the pixels being the same color or equal intensity
- Two pixels p & q are connected if
  - *They are adjacent in some sense*
  - *If their gray levels satisfy a specified criterion of similarity*

Connectivity

V: Set of gray levels used to define the criterion of similarity

4-connectivity

If gray level $(p, q) \in V, and\ q \in N_4(p)$

Set of gray levels V = {1}

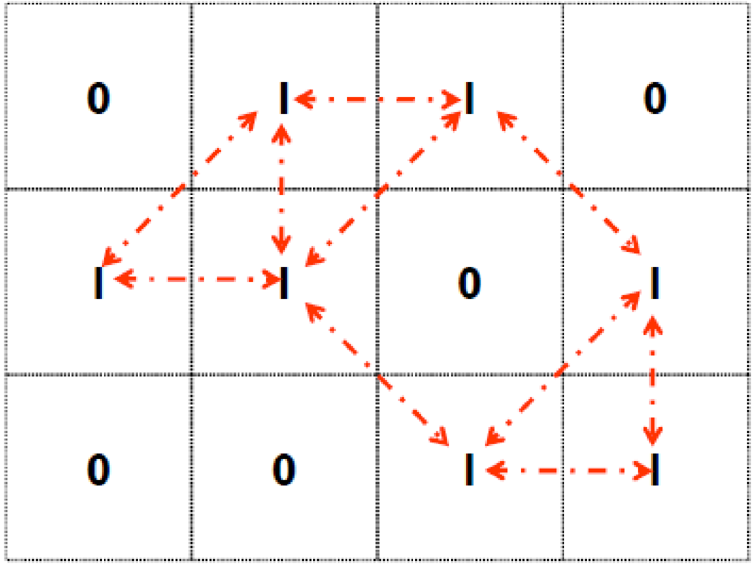
0	1	1	0
1	1	0	1
0	0	1	1

Connectivity

V: Set of gray levels used to define the criterion of similarity

8-connectivity

If gray level $(p,q) \in V, and\ q \in N_8(p)$



Set of gray levels V = {1}

Connectivity

V: Set of gray levels used to define the criterion of similarity

m-connectivity (Mixed Connectivity)

If gray level

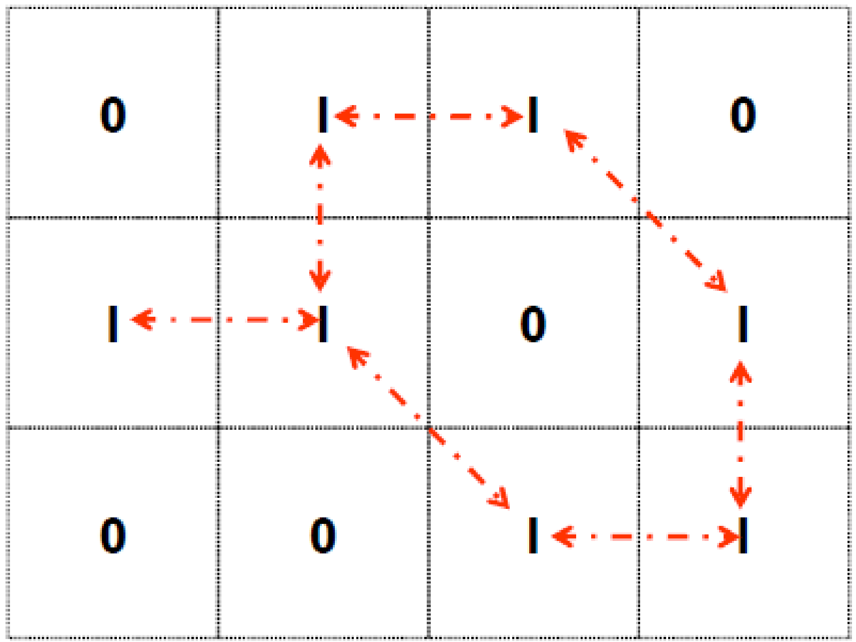
$(p,q) \in V, and\ q$ satisfies one of the following:

a. $q \in N_4(p)\ or$

b. $q \in N_D(p)\ And\ N_4(p) \cap N_4(q)$ has no pixels whose values are from V

Example: m – Connectivity

- Set of gray levels V = {1}



Note: Mixed connectivity can eliminate the multiple path connections that often occurs in 8-connectivity

Paths

- Path: Let coordinates of pixel *p: (x, y)*, and of pixel *q: (s, t)*
- A *path* from *p* to *q* is a sequence of distinct pixels with coordinates: $(x_0, y_0), (x_1, y_1), \ldots\ldots, (x_n,y_n)$

  where $(x_0, y_0) = (x, y)$ & $(x_n,y_n) = (s, t)$, *and* (x_i,y_i) *is adjacent to* (x_{i-1},y_{i-1}) $1 \leq i \leq n$

Test Yourself

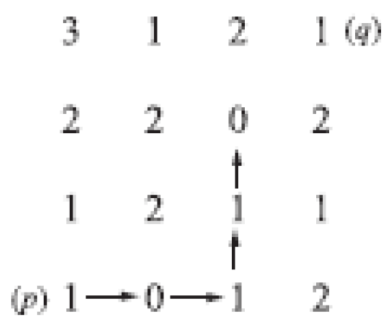
Consider the image segment shown.

(a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m-path between p and q. If a particular path does not exist between these two points, explain why.

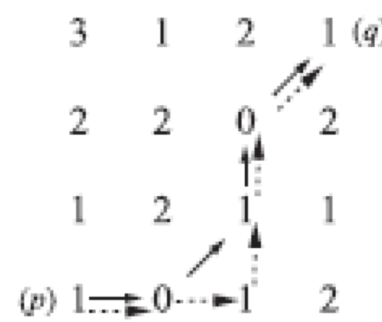
(b) Repeat for $V = \{1, 2\}$.

$$
\begin{array}{ccccc}
 & 3 & 1 & 2 & 1\,(q) \\
 & 2 & 2 & 0 & 2 \\
 & 1 & 2 & 1 & 1 \\
(p) & 1 & 0 & 1 & 2
\end{array}
$$

- When $V = \{0,1\}$, *4*-path does not exist between p and q because it is impossible to get from p to q by traveling along points that are both 4-adjacent and also have values from V . Fig. *a* shows this condition; it is not possible to get to q.
- The shortest *8*-path is shown in Fig. *b* its length is **4**.
- The length of the shortest *m*- path (shown dashed) is **5**.
- Both of these shortest paths are unique in this case.



(a)



(b)

Distance Metrics

- Let pixels *p*, *q* and *z* have coordinates *(x,y)*, *(s,t)* and *(u,v)* respectively.

- D is a distance function or metric if
  - $D(p,q) \geq 0$ and
  - $D(p,q) = 0$ iff $p = q$ and
  - $D(p,q) = D(q,p)$ and
  - $D(p,z) \leq D(p,q) + D(q,z)$

City block distance (D_4 distance)

$$D_4(p,q) = |x-s| + |y-t|$$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

- Diamond with center at *(x,y)*
- $D_4 = 1$ are the 4 neighbors of pixel *p(x,y)*

Chessboard distance (D_8 distance)

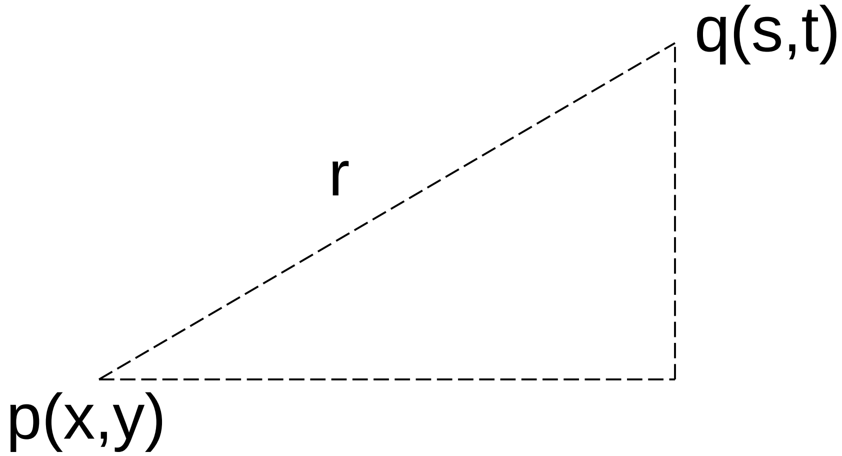
$$D_8(p,q) = \max(|x-s|, |y-t|)$$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

- Square centered at *p(x,y)*
- $D_8 = 1$ are the 8 neighbors of pixel *p(x,y)*

Euclidean Distance

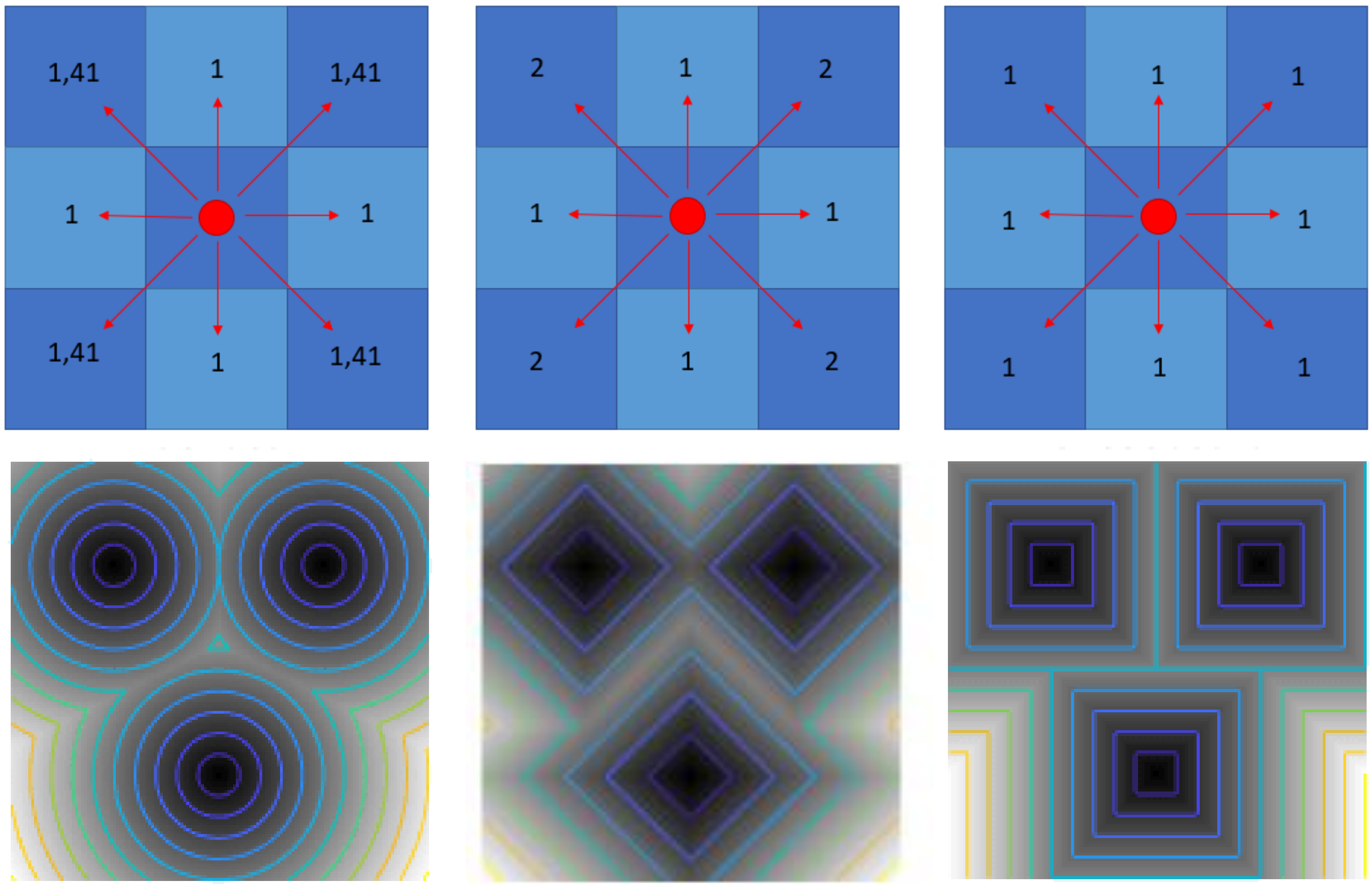
$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$



A circle with radius *r* centered at *(x,y)*

Distance Maps

EUCLIDEAN (STRAIGHT LINE) **L1 (CITY BLOCK)** **CHEBYSHEV (CHESSBOARD)**

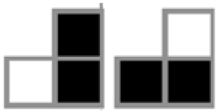


Connected Component Analysis

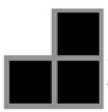
CC labeling – 4 Connectivity

- Process the image from left to right, top to bottom:

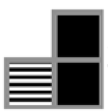
  1.) If the next pixel to process is 1



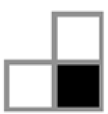
  i.) If only one of its neighbors (top or left) is 1, copy its label.



  ii.) If both are 1 and have the same label, copy it.

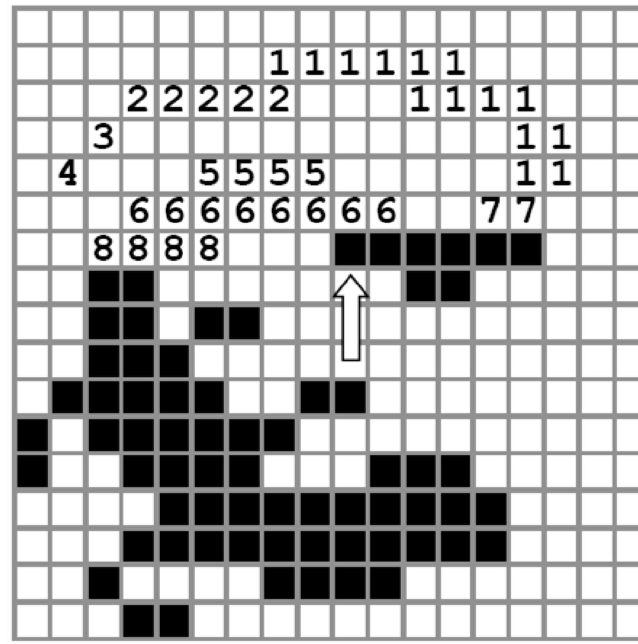


  iii.) If they have different labels
  - Copy the label from the left.
  - Update the equivalence table.



  iv.) Otherwise, assign a new label.

Pass 1

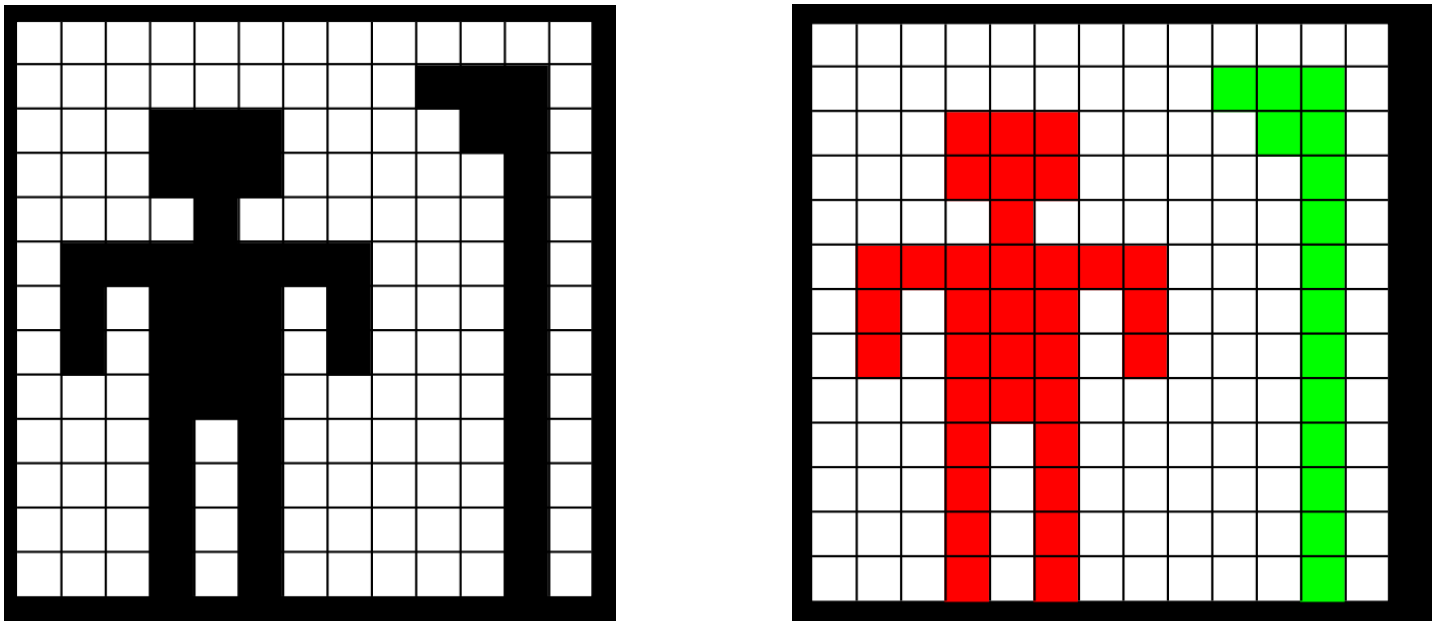


{1, 2, 7}
{3}
{4}
{5, 6, 8}
{}

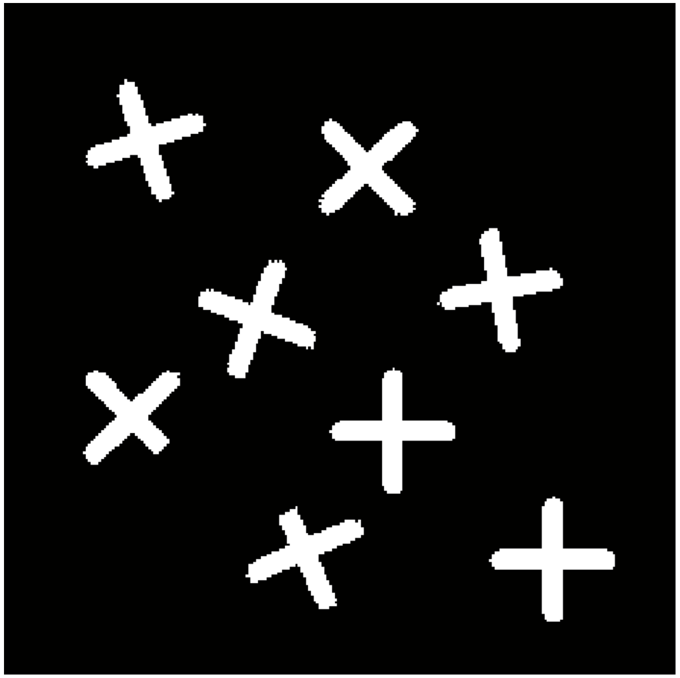
- Re-label with the smallest of equivalent labels

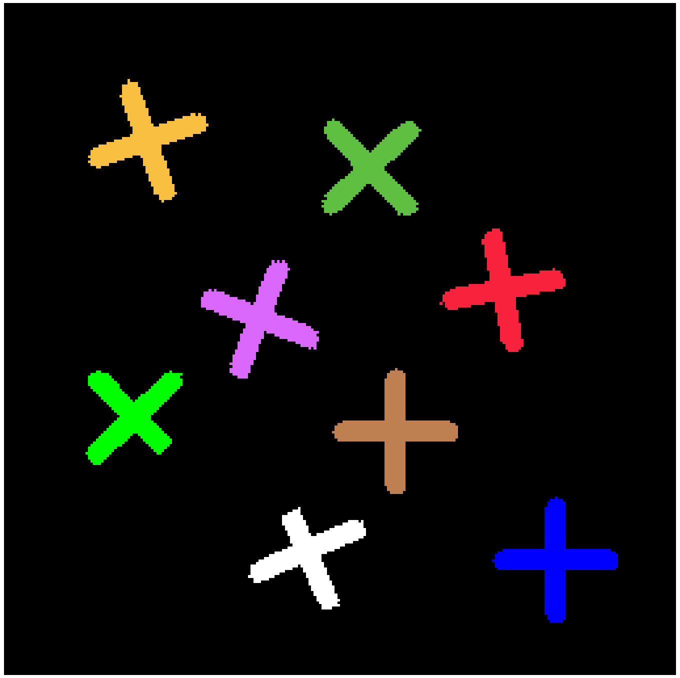
Pass 2

CC labeling – 4 Connectivity



CC labeling – 4 Connectivity

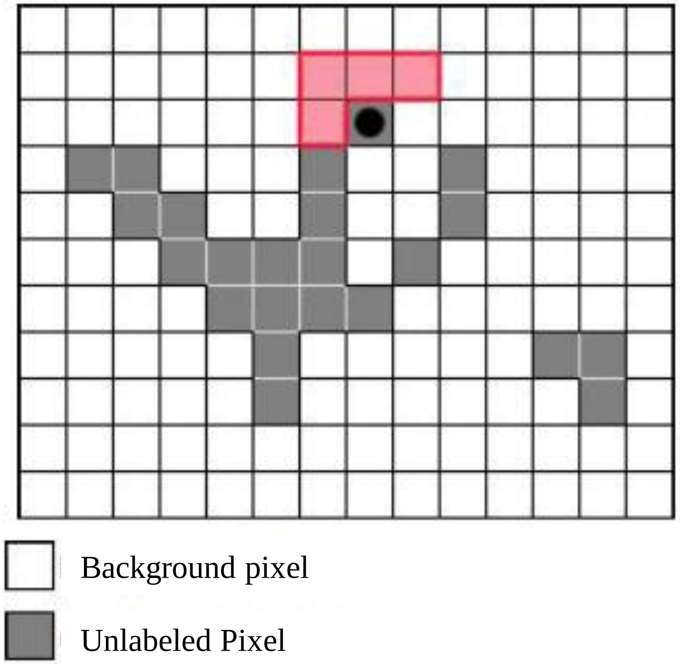


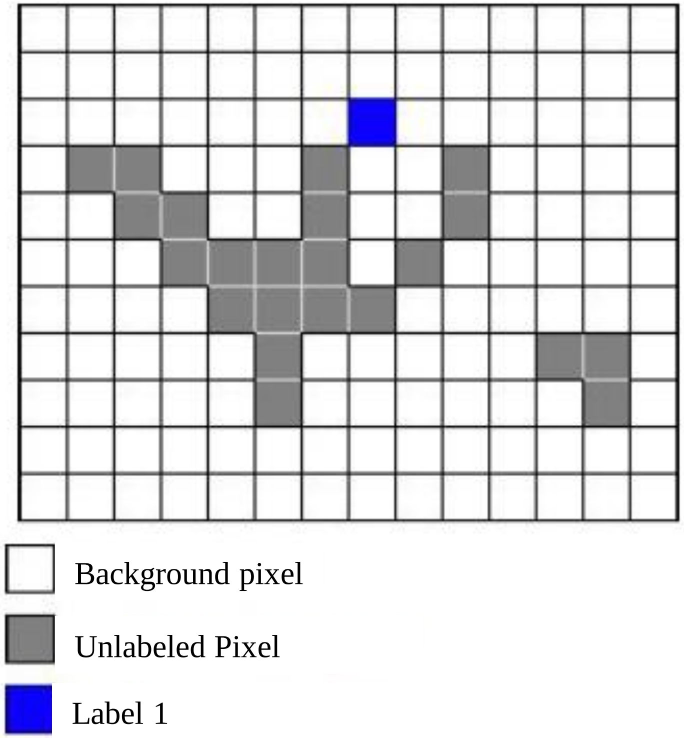


CC labeling – 8 Connectivity

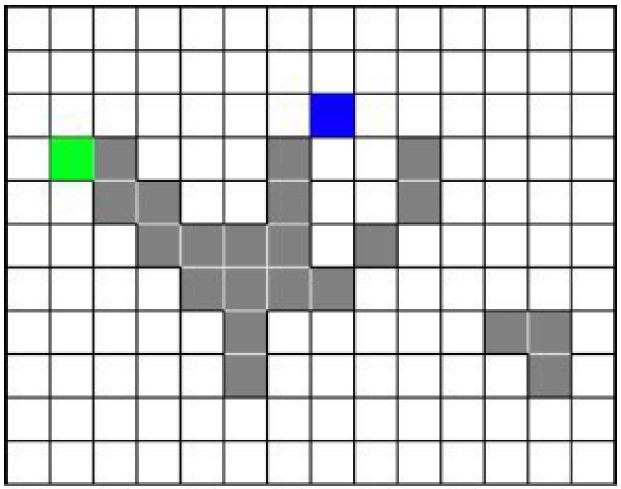
Same algorithm but examine also the upper diagonal neighbors of *p*

CC labeling – 8 Connectivity





CC labeling – 8 Connectivity

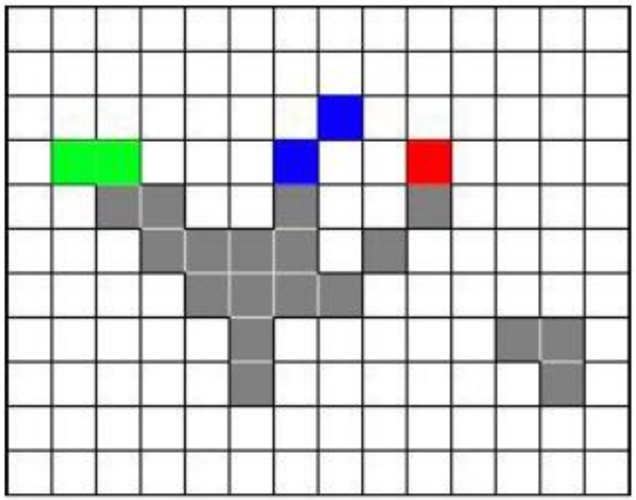


Background pixel

Unlabeled Pixel

Label 1

Label 2



Background pixel

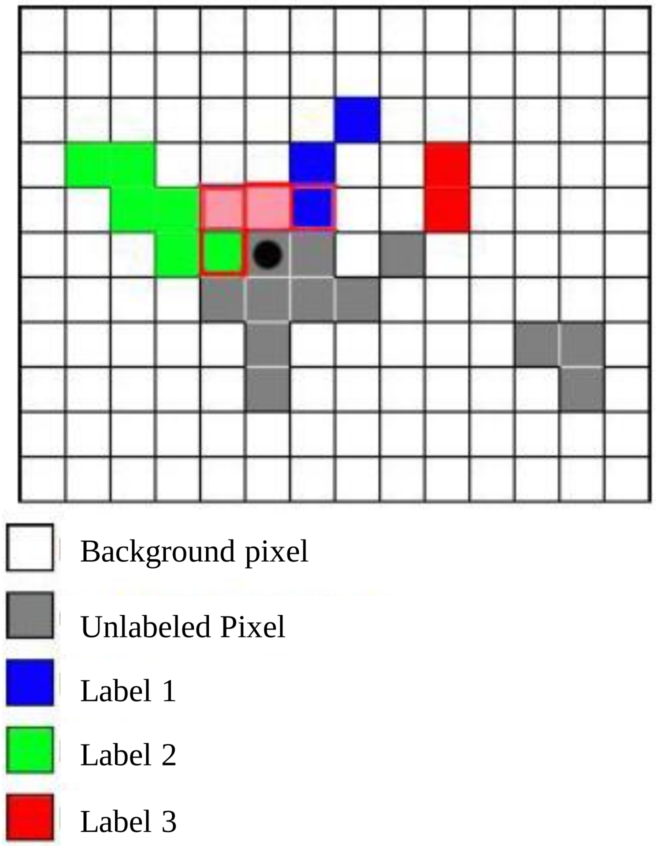
Unlabeled Pixel

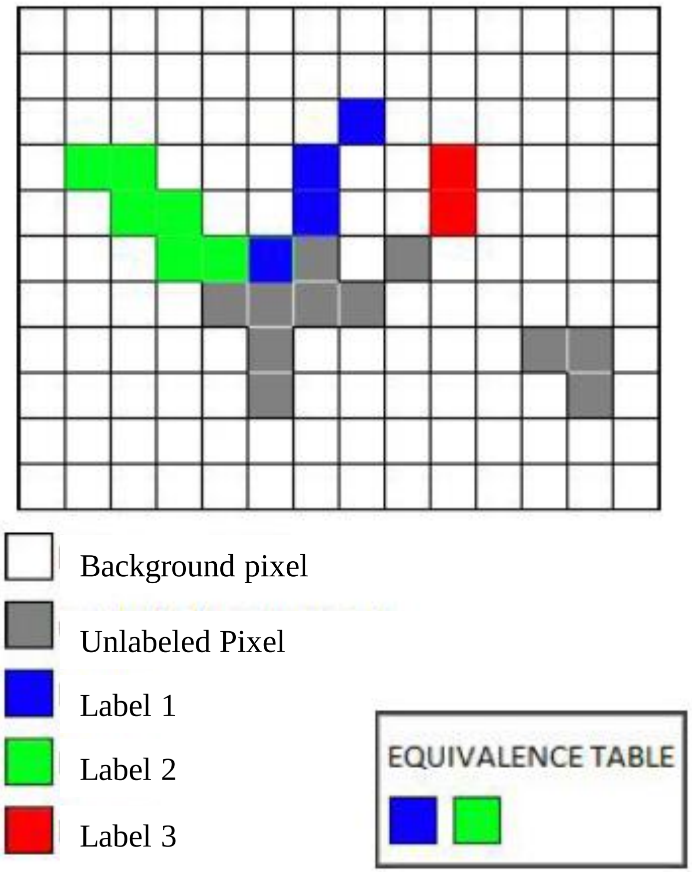
Label 1

Label 2

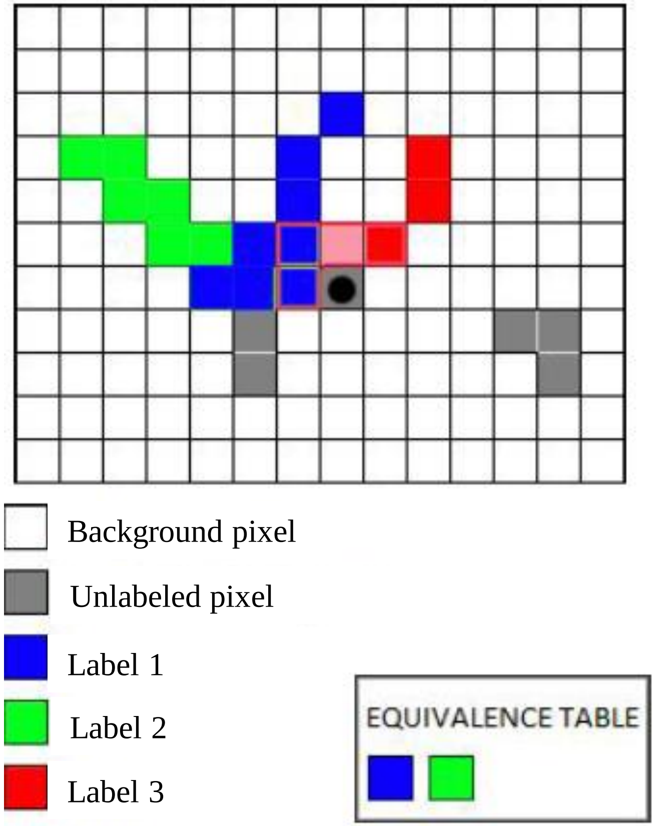
Label 3

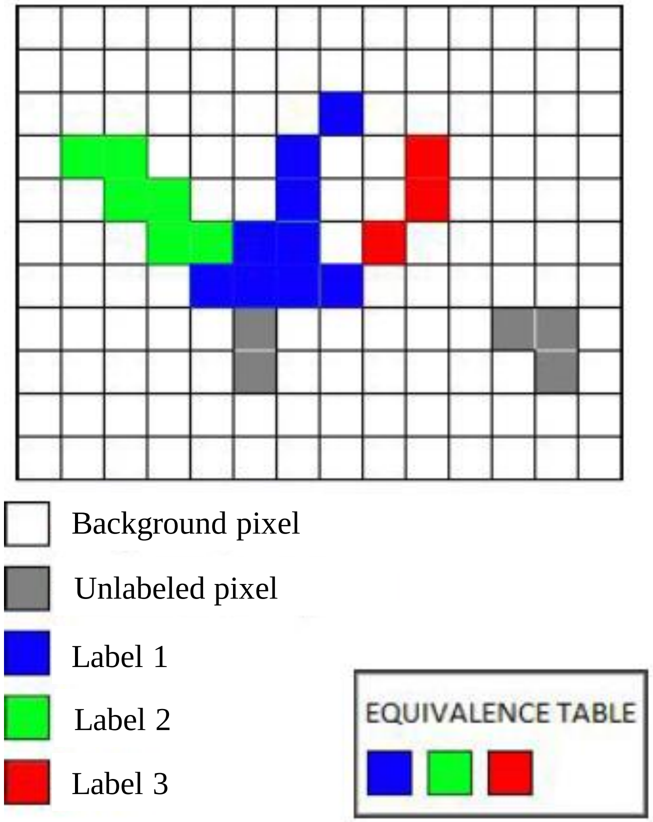
CC labeling – 8 Connectivity



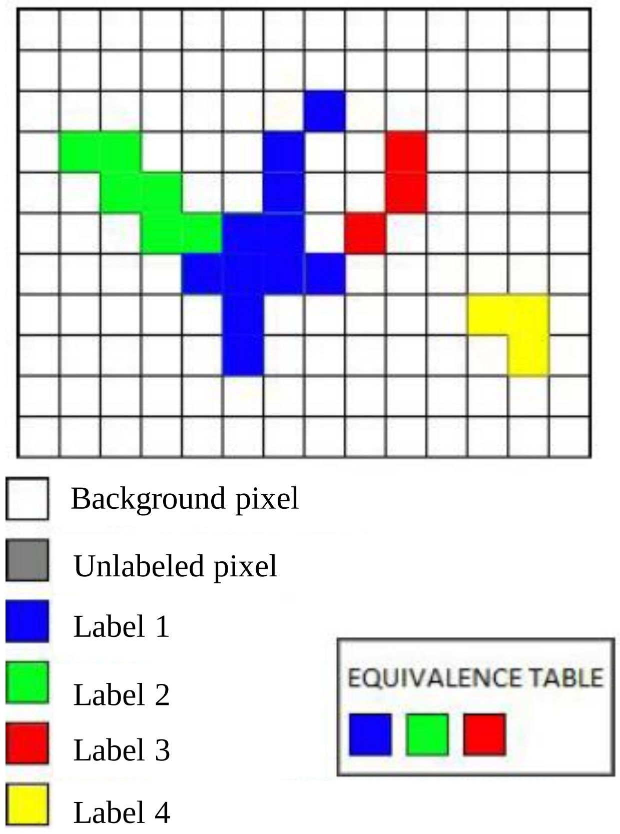


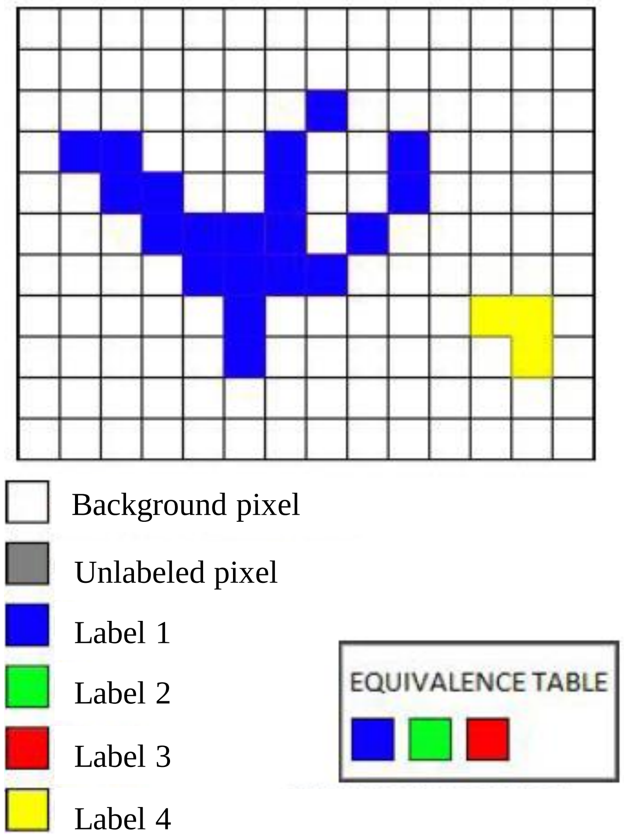
CC labeling – 8 Connectivity





CC labeling – 8 Connectivity





Readings from Book (3rd Edn.)

- Chapter – 2

Acknowledgements

Material in these slides has been taken from, the following resources

- Statistical Pattern Recognition: A Review – A.K Jain et al., PAMI (22) 2000
- Pattern Recognition and Analysis Course – A.K. Jain, MSU
- *Pattern Classification"* by Duda et al., John Wiley & Sons.
- Digital Image Processing", Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- Machine Vision: Automated Visual Inspection and Robot Vision", David Vernon, Prentice Hall, 1991
- www.eu.aibo.com/
- Advances in Human Computer Interaction, Shane Pinder, InTech, Austria, October 2008